3.4 Chemical changes

Changes that occur with the formation of new substance with different chemical composition or transformation of a substance into another substance with the evolution or absorption of heat or light energy are termed as chemical changes. Rusting of iron, burning, curdling of milk, reaction of baking soda with lemon juice, fermentation are some examples of chemical changes.

Chemical changes are very important in our lives. All the new substances which we use in various fields of our life are produced as a result of chemical reactions. Some of the examples of the importance of chemical changes are given below:

i. Metals are extracted from their naturally occurring compounds called 'ores' by a series of chemical changes.
ii. Medicines are prepared by carrying out a chain of chemical changes.
iii. The materials such as plastics, soaps, detergents, perfumes, acids, bases, salts etc are all made by carrying out various types of chemical changes.
iv. Every new material is discovered by studying different types of chemical changes.

In addition to new products, the following may also accompany a chemical change:

- Heat, light or any other radiation may be given off or absorbed.
- Sound may be produced
- A change in smell may take place (or) a new smell may be given off.
- A colour change may take place.
- A gas may be formed.

Explosion of a firework is a chemical change. We know that such an explosion produces heat, light, sound and unpleasant gases that pollute the atmosphere. That is why we are advised not to play with fireworks.

When food gets spoiled, it produces a foul smell. Shall we call this change as a chemical change? ..... Discuss in the class. Give your reflections.

You must have noticed that a slice of an apple acquires a brown colour if it is not consumed immediately. Colour of the potato remains the same when stored in water but there is change in colour with the piece kept in air. Look at the cut brinjal kept in air. The change of colour in these cases is due to the formation of some new substances which you will learn in higher classes. Are these not chemical changes?

Try yourself

Cut a fresh slice of potato and brinjal and keep it away for sometime.



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